

Meshach Ando

Jacksonville, FL | 904-749-2799 | meshachando.apply@gmail.com | linkedin.com/in/meshachando

PROFESSIONAL SUMMARY

Civil and environmental engineering graduate with 2+ years of experience in municipal solid waste (MSW) management, landfill engineering and environmental systems modeling. Experienced in Python, ArcGIS Pro, Microsoft Excel and environmental data analytics. Passed FE Civil exam and scheduled to take the PE Civil: Water Resources and Environmental exam on June 23, 2026.

EDUCATION

University of North Florida, Jacksonville, FL Aug. 2024 – May 2026

M.S., Civil Engineering

Kwame Nkrumah University of Science and Technology (KNUST), Kumasi Aug. 2018 – Nov. 2022

BSc. Civil Engineering

PROFESSIONAL EXPERIENCE

Graduate Research Assistant Aug. 2024 – May 2026

University of North Florida

Jacksonville, FL

- Compared fugitive methane (CH₄) emissions estimated using state-of-the-practice landfill emissions models (GHGRP, IPCC, SWICS, CARB and Dutch Afvalzorg) against direct measurements from 313 MSW landfills across 42 U.S. states.
- Developed a Python-based inverse modeling framework for estimating whole-landfill CH₄ emissions from landfill surface emissions monitoring (SEM) data.
- Assessed PFAS exposure risks around MSW landfills at occupational, public health and ambient receptor distances.

Staff Engineer Nov. 2022 – Nov. 2023

Kumasi Compost and Recycling Plant (KCARP)

Kumasi, Ghana

- Conducted “sampling and sorting” waste characterization studies to determine received waste composition and evaluate sorting plant efficiency.
- Collected and analyzed waste generation, recovery and disposal data to evaluate sorting plant performance and residual waste landfill operations.
- Developed the “KCARP–Kumasi SSaRP” sanitation-behavioral model to predict recycling participation and source-separation practices among college students in Kumasi.
- Supported construction and environmental evaluation of a 215 m × 250 m engineered residual solid waste landfill cell.

Environmental Engineering Intern Sep. 2021 – Dec. 2021

Zoomlion Ghana Limited

Obuasi, Ghana

- Monitored municipal solid waste collection operations in Service Zones A and F to support the daily supervision of approximately 290 sanitation personnel.
- Inspected the Kwabrafoso municipal solid waste landfill to evaluate waste placement practices, daily cover application and operational compliance.
- Assessed source-separation practices among residents of Kokoteasua through field surveys conducted as part of municipal waste management improvement initiatives.

PROJECTS

Comparative Evaluation of Landfill Methane Emission Models

Aug. 2024 – May 2026

Compared methane emissions estimated using GHGRP, IPCC, SWICS, CARB and Dutch Afvalzorg landfill models against direct measurements from municipal solid waste landfills across the United States.

Surface Emissions Monitoring-to-Emission Rate Framework

Aug. 2024 – May 2026

Developed a Python-based inverse modeling framework for estimating whole-landfill methane emission rates and source locations from landfill surface emissions monitoring datasets.

PFAS Risk Assessment Study

Aug. 2024 – May 2026

Dispersion modeling to estimate PFAS concentrations to assess occupational and public health risks associated with volatile PFAS exposure.

CERTIFICATIONS

FE Civil – Passed

Feb. 2026

PE Civil: Water Resources and Environmental Exam Scheduled

June 23, 2026

SOFTWARE AND DATA ANALYTICS

Programming and Software: Python (Advanced), ArcGIS Pro (Advanced), Microsoft Excel (Advanced), AutoCAD (Moderate), Civil 3D (Beginner)

Data Analytics and Modeling: Environmental data analytics, statistical analysis, GIS analysis, inverse modeling, uncertainty analysis, environmental systems modeling

PRESENTATIONS

Poster Presentation – Association of Environmental Engineering and Science Professors (AEESP) Conference

Mar. 2025

Title: Environmental Systems Modeling and Landfill Emissions Research

Poster Presentation – Global Waste Management Symposium (GWMS)

Mar. 2026

Title: Modeled and Measured Landfill Methane Emissions

PREPARED PUBLICATIONS AND MANUSCRIPTS

Meshach Ando, Andres Leiva and Florentino B. De la Cruz. **Comparison of Field Measurements to Methane Emissions Models at Municipal Solid Waste Landfills in the United States.** *Formatted for submission to Environmental Science & Technology.* Manuscript in preparation.

Andrés Leiva-Araos, Meshach Ando, José García and Florentino B. de la Cruz. **Generalizable Cross-Landfill Methane Estimation in Waste Management Using Machine Learning and Metaheuristics.** *Formatted for submission to Elsevier Soft Computing.* Manuscript in preparation.

Florentino B. De la Cruz, Silas Buckner, John Nuszowski, Meshach Ando, Andrés Leiva Araos, Jamie DeWitt, Ivan A. Titaley, Jennifer A. Field and Morton A. Barlaz. **Are Volatile Per- and Polyfluoroalkyl Substances (PFAS) Attenuated in Landfill Cover?** *Formatted for submission to Engineering Science and Technology.* Manuscript in preparation.